SiteXRD-Bench: A Hierarchical Benchmark for Chemical-Order Inference and Overlapping-Peak Decomposition in Structural Alloys

Anonymous Author(s)

Abstract

Powder X-ray diffraction (XRD) is a central materials characterization signal, but it is also a lossy inverse problem: many three-dimensional atomic states can map to nearly the same one-dimensional pattern. Existing XRD machine-learning benchmarks largely reward phase labels or whole-pattern reconstruction, leaving open a harder question: did the model recover the latent physical state that generated the observation?

We introduce SiteXRD-Bench, one benchmark with two independent task tracks for latent-state recovery from powder XRD. HEA-SRO provides 84,019 quality-controlled high-entropy-alloy structure–XRD pairs with permutation-invariant first- and second-shell Warren–Cowley short-range order targets under ID, composition OOD, chemistry OOD, SRO OOD, and matched simulated perturbations. Gamma-Decomp (gamma/gamma-prime decomposition) provides 10,000 γ/γ' mixture patterns and 21,116 overlap windows with pre-mixing phase/HKL component truth, peak centers, widths, and areas.

The benchmark exposes both learnable structure and principled non-identifiability. On ID HEA-SRO, Spectrum Transformer improves SRO MAE over Composition Ridge by 24.4% (95% group-bootstrap interval 23.8–24.9%). On Gamma-Decomp, Peak Feature improves component F1 over a no-spectrum prior by 0.260 (0.207–0.313) on full patterns and 0.740 (0.717–0.762) on overlap windows. Yet none of 4,000 same-composition occupancy pairs exceeds the frozen simulated-perturbation identifiability threshold. Thus SiteXRD-Bench evaluates not only accuracy, but also shortcut use, OOD behavior, component ambiguity, and abstention. Because no repeated experimental spectra have exact SRO or component truth, v1.1 makes no claim of experimental-noise calibration or unique occupancy recovery.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Mathematics of computing** → *Probability and statistics.*

Keywords

X-ray diffraction, short-range order, high-entropy alloys, Ni-based superalloys, inverse problems, out-of-distribution generalization

ACM Reference Format:

Anonymous Author(s). 2027. SiteXRD-Bench: A Hierarchical Benchmark for Chemical-Order Inference and Overlapping-Peak Decomposition in Structural Alloys. In *Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '27).* ACM, New York, NY, USA, 7 pages.

1 Introduction

Powder XRD is non-destructive, scalable, and sensitive to long-range crystal structure, but its inverse problem is underdetermined: a measured one-dimensional pattern aggregates many reflections and can be compatible with multiple latent structures. This matters when the target is local chemical order or phase-resolved components rather than a coarse phase label.

The motivating failure case is concrete. In the public HEA training split, two $Co_{0.25}Cr_{0.25}Fe_{0.25}Ni_{0.25}$ FCC structures selected by a frozen rule have XRD cosine similarity 0.9967 but first/second-shell SRO L_1 distance 13.33 (Figure 2). A model can fit the dominant Bragg pattern and still miss the local chemical order. The same disconnect appears in γ/γ' mixtures: a low residual after summing peaks does not show whether intensity was assigned to the correct phase/HKL components. SiteXRD-Bench is built around this distinction between looking right and being physically right.

Two alloy settings expose complementary ambiguities. In high-entropy alloys (HEAs), structures with the same composition and BCC/FCC parent lattice can differ in atomic arrangement and short-range order (SRO), while their dominant Bragg peaks remain similar [4, 18, 25]. In Ni-based superalloys, FCC γ and $L1_2$ γ' have related lattices; their reflections overlap under broadening, strain, and acquisition effects [3, 17, 20].

SiteXRD-Bench separates observational fit from latent correctness. It contributes (1) a shared structure–reflection–component–observation hierarchy, (2) same-composition HEA controls and traceable γ/γ' mixtures, (3) task-specific permutation-invariant metrics, and (4) grouped ID/OOD protocols, identifiability diagnostics, and parent-aware uncertainty estimates. Strength and creep-life prediction are outside the benchmark contract.

2 Related Work and Positioning

Rietveld refinement remains the canonical whole-pattern framework [21]; pseudo-Voigt line shapes provide a standard representation of diffraction broadening [22]. A good profile fit, however, does not prove unique site occupancy. Warren–Cowley coefficients summarize conditional neighbor probabilities without assigning arbitrary atom indices [2].

XRD learning has addressed symmetry classification, augmentation, and multiphase identification or quantification [6, 12, 19, 23], while set prediction supports unordered component outputs [1, 11]. Calibration and domain generalization motivate controlled shifts rather than random-split accuracy alone [7, 8].

3 Dataset Overview

For latent material state z and acquisition condition n, the release stores

$$z \longrightarrow \mathcal{R}(z) \longrightarrow \{c_k(\theta; z, n)\}_{k=1}^{K} \longrightarrow y(\theta), \tag{1}$$

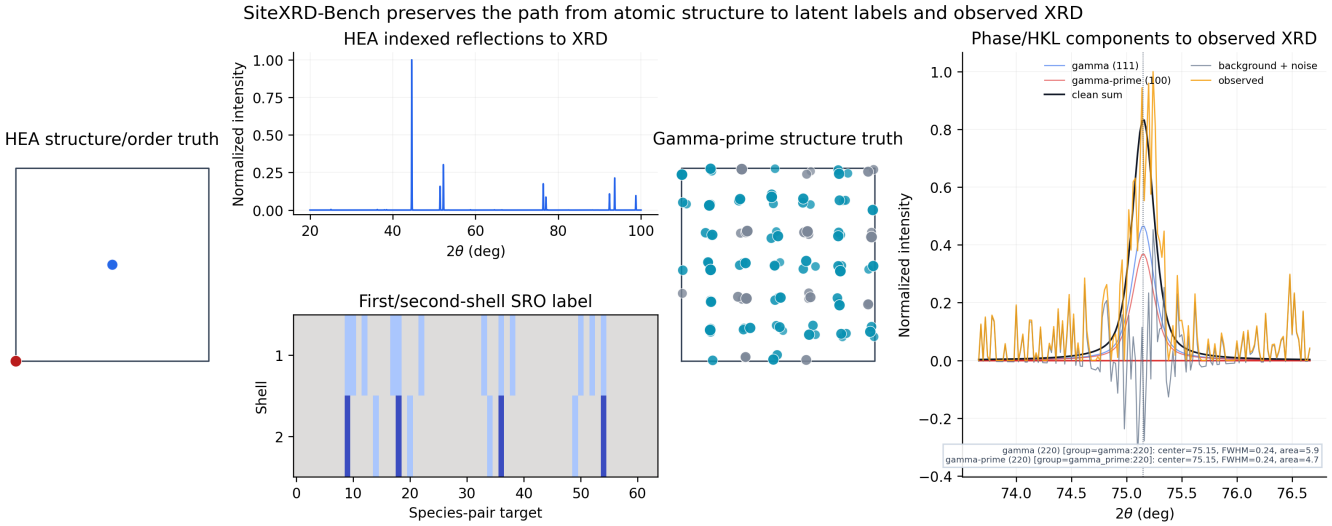


Figure 1: The two released data tracks and their traceable forward chains. Every displayed object comes from a public training manifest; IDs, selection metrics, reasons, and input hashes are stored in the visual manifest.

Table 1: Representative resources and released supervision. "Missing" denotes absence from that resource's contract, not a defect.

Resource	Released evidence	Missing for this benchmark
COD [5]	Open crystal structures	Component truth and grouped inverse evaluation
Materials Project [9]	Computed structures/properties	Observed-pattern component provenance
RRUFF [10]	Experimental mineral XRD	SRO and phase/HKL component truth
HEA source [14]	Relaxed structures/attributes	XRD tasks, perturbations, evaluator
SiteXRD-Bench	SRO, reflections, components, XRD, OOD	Exact experimental latent truth

where $\mathcal{R}$ is the indexed reflection set and c_k is a broadened phase/HKL component. The observation is

$$y(\theta) = \sum_{k=1}^{K} c_k(\theta; z, n) + b(\theta; n) + \epsilon(\theta; n). \tag{2}$$

Stable sample, structure-state, parent-group, reflection, and component IDs link every level to a frozen manifest, configuration, checksum, and provenance row.

Every source row receives a stable ID, canonical structure hash, composition, quality result, and eligibility flags. Derived labels are deterministic, versioned functions of formal structures. Rejected rows remain accounted for. Perturbed views inherit their clean parent's group and cannot cross a split. Thus reconstruction can be scored independently from recovery of z or $\{c_k\}$.

4 HEA-SRO Data

The source corpus contains ordered cells and SQS structures with up to 125 atoms from Al, Co, Cr, Cu, Fe, Mn, Ni, and Si [13, 14]. Of 84,024 source rows, 84,019 pass quality checks and five are rejected. Each accepted structure has one immutable clean Cu Kα pattern with 4,001 points. Perturbations are indexed as views, not additional structures.

For species a, b and shell s, the primary target is

$$\alpha_{ab}^{(s)} = 1 - \frac{P(b \mid a, s)}{c_b}, \tag{3}$$

where c_b is the global concentration. Each sample stores complete masked first/second-shell species-pair matrices, topology, shell definitions, and provenance to the relaxed structure. Exact atom-index recovery is not a task.

Matching composition and topology blocks a composition shortcut, while the near-identical spectra retain different local order. This is why the dataset needs both SRO labels and identifiability analysis. Family is retained only as an automated evidence diagnostic. A 1,000-parent counterfactual diagnostic contains 4,000 same-composition occupancy pairs; none exceeds its parent-specific simulated-noise floor. Ambiguous pairs are not ranking negatives, and the threshold is not lowered to meet a quota.

5 Gamma/Gamma-Prime Decomposition Data

Gamma-Decomp is the stable task ID for this γ/γ' track. The generator pairs FCC γ and $L1_2$ γ' structures and varies phase weight, broadening, strain, crystallite size, background, zero shift, and noise within declared ranges. Physically relaxed and parameterized counterfactual pairs are explicitly distinguished. Component truth is stored before summation.

Gamma-Decomp v1.1 contains 10,000 mixtures, 21,116 windows, 175,150 raw reflection-provenance rows, and 165,150 full-pattern scoring components. The window view has 34,983 scoring components. Strictly coincident γ' (221) and (300) reflections remain separate in provenance but merge into an equivalent HKL scoring group. Seventeen phase/HKL groups are evaluated.

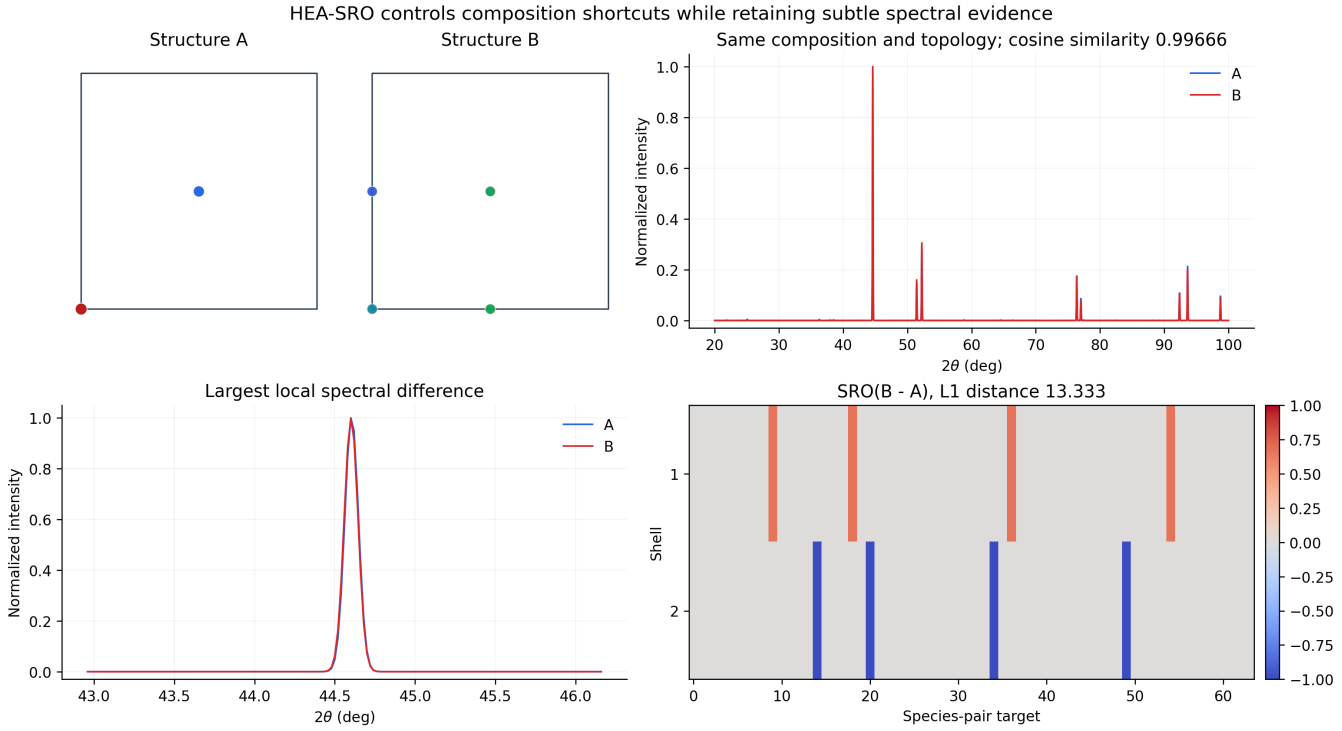


Figure 2: Deterministically selected HEA-SRO training pair: same reduced composition and BCC/FCC topology, XRD cosine similarity at least 0.995, and maximum eligible SRO L_1 difference. The full spectra, most different local peak region, and first/second-shell SRO difference are shown.

Overlap difficulty is quantitative and frozen in the manifest. For every pair of components in a window, the separation ratio is

$$\rho_{ij} = |m_i - m_j| / \{(w_i + w_j)/2\},$$

where m is peak center and w is FWHM. The window difficulty uses $\rho_{\min} = \min_{i<j} \rho_{ij}$: single if no scored pair exists, severe if $\rho_{\min} < 0.5$, strong if $0.5 \le \rho_{\min} < 1$, partial if $1 \le \rho_{\min} < 2$, and resolved if $\rho_{\min} \ge 2$. Thus "severe overlap" means peak centers are separated by less than half the average FWHM, not an aesthetic choice made after seeing model errors.

The target is an unordered component set with phase, equivalent HKL group, center, width, area, and profile parameters. An observed peak may contain several such components. Low reconstruction error therefore does not imply correct latent decomposition; evaluation uses deterministic optimal bipartite matching rather than output order.

6 Coverage, Tasks, and Splits

The clean reference uses Cu Kα, 20°–100° in 2θ, and 0.02° spacing. Reflection tables retain position, intensity, HKL, and multiplicity. Clean truth is separated from broadening, background, strain, texture, and noise. Validation/test perturbations are frozen.

HEA groups structure hashes, parents, and all observation views before assigning ID, composition OOD, chemistry OOD, and SRO OOD splits. Gamma groups parent pairs and views before instrument and overlap slices. Nearest-neighbor audits cover structure and spectral spaces. An ordinary random split is not presented as evidence of new-chemistry or acquisition generalization.

HEA macro-averages masked MAE within a parent and then across parents. Relative improvement over mandatory Composition Ridge is

$$S_{\mathrm{SRO}} = 100 \max(0, 1 - \mathrm{MAE}_{model}/\mathrm{MAE}_{base}). \quad (4)$$

Gamma reports component F1, phase/HKL and count accuracy, and errors in center, width, area, and profile. Whole-pattern error remains supplementary. Random-seed standard deviations and 2,000-replicate paired-parent bootstrap intervals (seed 2027) are reported separately. No cross-track composite score is defined because SRO matrix regression and unordered component recovery have different target spaces and metrics.

7 Benchmark Validation

The formal HEA matrix consists of Composition Ridge, Peak XGBoost, ResNet1D, and Spectrum Transformer. Ridge is deterministic; the three XRD models use split-specific training memberships and seeds 2027–2031. A physics-constrained HEA experiment is appendix-only and is not part of the formal four-model matrix. Gamma adds a no-spectrum prior, constrained pseudo-Voigt fit, peak features, neural set predictors, and a physics-constrained component model.

The physics-constrained variants are reported but not promoted into the primary leaderboard for two reasons. First, they introduce

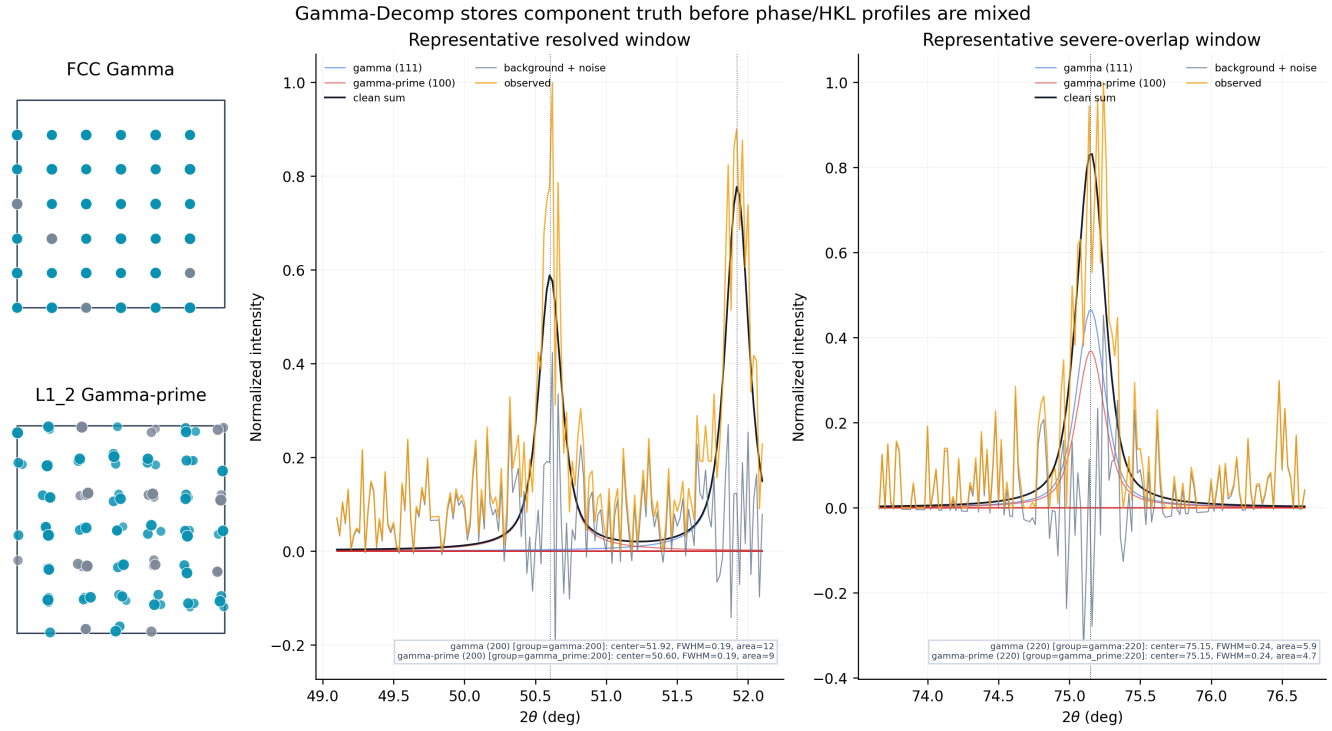


Figure 3: Gamma-Decomp training examples nearest to the resolved and severe-overlap medians in component count, SNR, and phase weight. The panels show FCC γ, $L1_2$ γ', phase/HKL profiles, clean sum, background/noise residual, observed pattern, center, FWHM, and area.

Table 2: Dataset and task contract. Both tracks belong to one release and are scored independently. Reconstruction is supplementary and never replaces a latent metric.

Track	Released scale	Target	Frozen splits	Primary metrics
HEA-SRO	84,019 structure–XRD pairs	Masked first/second-shell Warren–Cowley matrices	ID; composition, chemistry, SRO OOD	Parent-macro masked MAE; per-shell Spearman
Gamma-Decomp	10,000 mixtures; 21,116 windows	Unordered phase/HKL components and peak parameters	Instrument ID/OOD; resolved, partial, severe	Matched component F1; phase/HKL and count accuracy

additional forward-profile structure and latent peak-parameter priors, so they test a different contract than end-to-end learning from the released inputs. Second, their gains are not uniform: in HEA-SRO the appendix variant changes MAE by only 0.0011–0.0024 relative to Spectrum Transformer, while in Gamma-Decomp it is competitive on full patterns but much weaker on fixed overlap windows. Keeping them in the appendix makes the design choice explicit without letting a specialized prior define the benchmark's main comparison.

On HEA ID, Spectrum Transformer reduces parent-macro MAE from 0.4496 to 0.3400. The 24.4% reduction has a parent-bootstrap interval of 23.8–24.9% over 12,556 groups. Peak XGBoost is competitive; ResNet1D does not consistently beat the metadata control, demonstrating why a simple shortcut baseline is necessary.

On Gamma test data, Peak Feature attains component F1 of 0.683 on full patterns and 0.809 on windows, versus 0.421 and 0.072 for the no-spectrum prior. For seed 2027, paired-parent improvements are 0.260 (0.207–0.313) and 0.740 (0.717–0.762). Neural predictors are not uniformly superior: the Gamma physics-constrained model scores 0.627 on full patterns but 0.304 on windows. The occupancy diagnostic remains a formal negative result: 0/4,000 pairs exceed the frozen ratio 1.0.

All prediction IDs must match the declared split exactly. Perfect-oracle tests attain the exact optimum, shuffled predictions degrade, and component permutations leave set scores unchanged. Release readiness depends on complete IDs, splits, seeds, manifests, checksums, and statistics, not on whether a model beats Ridge.

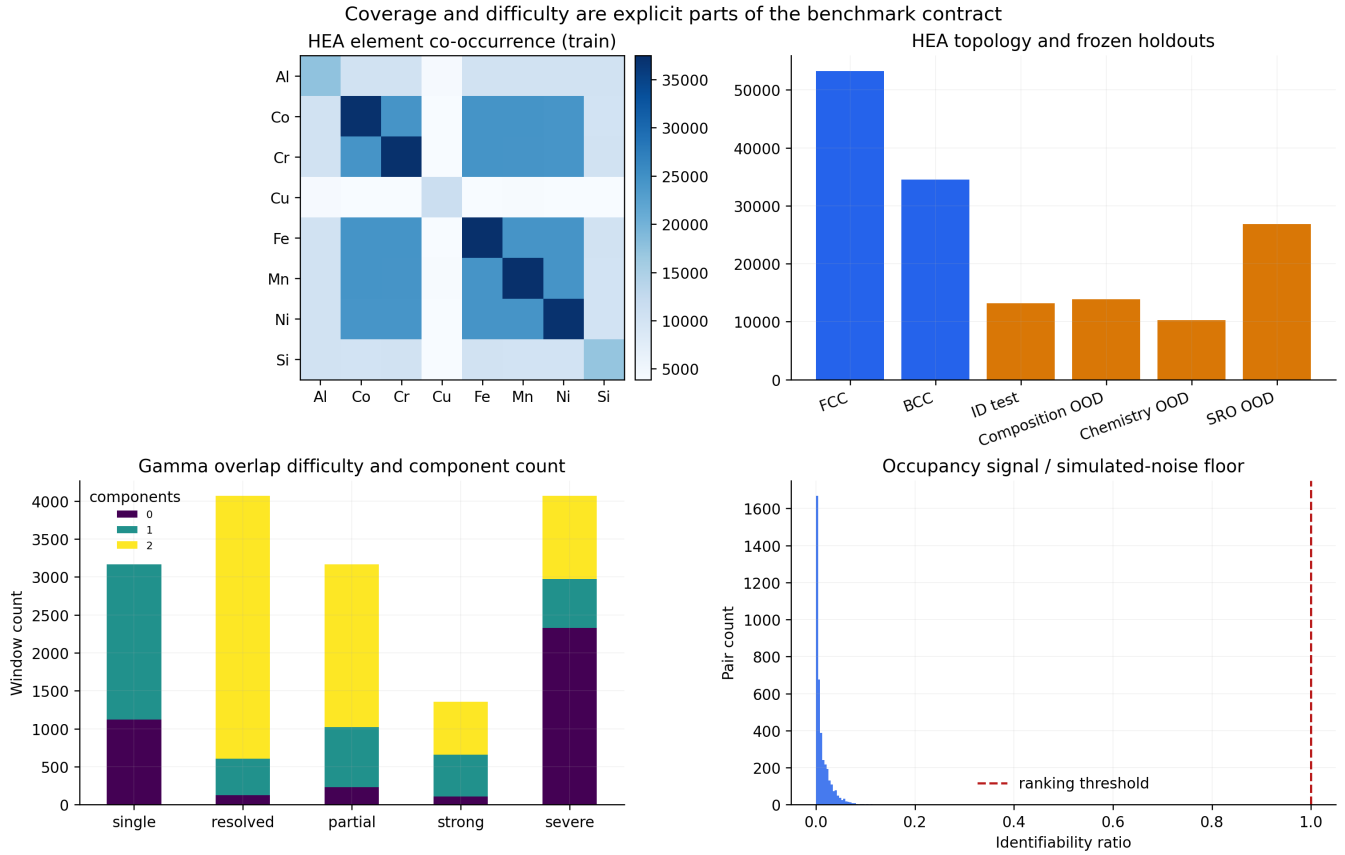


Figure 4: Manifest-derived coverage and difficulty: HEA element co-occurrence, BCC/FCC and ID/OOD support, Gamma component-count/overlap support, and occupancy identifiability ratio with frozen threshold 1.0.

Table 3: Compact primary results. HEA reports parent-macro SRO MAE for ID and three OOD splits; Gamma reports component F1 for full patterns and windows. SD is across declared seeds; "–" means not applicable.

HEA-SRO parent-macro MAE

Split	Model	MAE ↓	SD
chemistry_ood	Composition Ridge	0.4340	0.0000
chemistry_ood	Peak XGBoost	0.3575	0.0000
chemistry_ood	ResNet1D	0.4396	0.0055
chemistry_ood	Spectrum Transformer	0.3362	0.0035
composition_ood	Composition Ridge	0.4433	0.0000
composition_ood	Peak XGBoost	0.3621	0.0000
composition_ood	ResNet1D	0.4632	0.0164
composition_ood	Spectrum Transformer	0.3402	0.0022
id	Composition Ridge	0.4496	0.0000
id	Peak XGBoost	0.3621	0.0000
id	ResNet1D	0.4414	0.0202
id	Spectrum Transformer	0.3400	0.0009
sro_ood	Composition Ridge	0.5142	0.0000
sro_ood	Peak XGBoost	0.4493	0.0000
sro_ood	ResNet1D	0.5020	0.0136
sro_ood	Spectrum Transformer	0.4291	0.0025

Gamma-Decomp component F1

View	Model	F1 ↑	SD
Full	No-spectrum prior	0.421	–
Window	No-spectrum prior	0.072	–
Full	Peak Feature	0.683	0.003
Window	Peak Feature	0.809	0.002
Window	Pseudo-Voigt (BIC)	0.527	–
Full	ResNet Set	0.473	0.015
Window	ResNet Set	0.466	0.114
Full	Transformer Set	0.599	0.163
Window	Transformer Set	0.494	0.038
Full	Physics-constr.	0.627	0.155
Window	Physics-constr.	0.304	0.059

8 Availability, Limitations, and Conclusion

Code, configuration, evaluation, cards, download tools, and smoke data are released separately from the sharded structures/XRD/labels and the frozen metadata/checksum snapshot. The final paper records public GitHub, Hugging Face, and Zenodo DOI links only after the records exist and anonymous HTTP checks succeed; no provisional URL or unminted DOI is printed. The repository also contains the Dataset Card, Benchmark Card, Croissant metadata, citation record, and release checklist.

The computational structure is ground truth for generated samples, but does not prove that experimental powder XRD uniquely

determines occupancy. Periodic supercells, similar scattering factors, finite resolution, texture, microstrain, nanostructure, and unmodeled phases can make inversion ambiguous. Family labels are automated and unadjudicated. Occupancy uses simulated perturbations only; real repeated-measurement import and noise calibration are v1.2 gates.

The admitted HEA experimental set supports only mother-phase trends [24]. The Ni-based sanity set contains licensed raw arrays for heat-treated IN718 and AM718R, where γ' and gamma-double-prime superlattice reflections are not distinguishable [15, 16]. It supports phase-level and whole-pattern checks, not experimental component-level validation.

SiteXRD-Bench therefore unifies two complementary inverse problems within one versioned benchmark: subtle local chemical order and overlapping γ/γ' phase evidence. Its central claim is not that every latent variable is recoverable, but that recoverability, shortcut use, OOD behavior, and abstention can now be evaluated against traceable truth.

References

[1] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. 2020. End-to-End Object Detection with Transformers. In *Computer Vision – ECCV 2020*. Springer, Cham, Switzerland, 213–229. doi:10.1007/978-3-030-58452-8_13

[2] J. M. Cowley. 1950. An Approximate Theory of Order in Alloys. *Physical Review* 77, 5 (1950), 669–675. doi:10.1103/PhysRev.77.669

[3] A. Devaux, L. Nazé, R. Molins, A. Pineau, A. Organista, J. Y. Guédou, J. F. Uginet, and P. Héritier. 2008. Gamma Double Prime Precipitation Kinetic in Alloy 718. *Materials Science and Engineering: A* 486, 1–2 (2008), 117–122. doi:10.1016/j.msea.2007.08.046

[4] Easo P. George, Dierk Raabe, and Robert O. Ritchie. 2019. High-Entropy Alloys. *Nature Reviews Materials* 4, 8 (2019), 515–534. doi:10.1038/s41578-019-0121-4

[5] Saulius Gražulis, Adriana Daškevič, Andrius Merkys, Daniel Chateigner, Luca Lutterotti, Miguel Quirós, Natalia R. Serebryanaya, Peter Moeck, Robert T. Downs, and Armel Le Bail. 2012. Crystallography Open Database (COD): An Open-Access Collection of Crystal Structures and Platform for World-Wide Collaboration. *Nucleic Acids Research* 40, D1 (2012), D420–D427. doi:10.1093/nar/gkr900

[6] Jaimie Greasley and Patrick Hosein. 2023. Exploring Supervised Machine Learning for Multi-Phase Identification and Quantification from Powder X-Ray Diffraction Spectra. *Journal of Materials Science* 58, 12 (2023), 5334–5348. doi:10.1007/s10853-023-08343-4

[7] Ishaan Gulrajani and David Lopez-Paz. 2021. In Search of Lost Domain Generalization. In *International Conference on Learning Representations*. OpenReview.net.

[8] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning*. PMLR, 1321–1330.

[9] Anubhav Jain, Shyue Ping Ong, Geoffroy Hautier, Wei Chen, William Davidson Richards, Stephen Dacek, Shreyas Cholia, Dan Gunter, David Skinner, Gerbrand Ceder, and Kristin A. Persson. 2013. Commentary: The Materials Project: A Materials Genome Approach to Accelerating Materials Innovation. *APL Materials* 1, 1 (2013), 011002. doi:10.1063/1.4812323

[10] Barbara Lafuente, Robert T. Downs, Hexiong Yang, and Nate Stone. 2015. The Power of Databases: The RRUFF Project. In *Highlights in Mineralogical Crystallography*, Thomas Armbruster and Rosa Micaela Danisi (Eds.). De Gruyter, 1–30. doi:10.1515/9783110417104-003

[11] Juho Lee, Yoonho Lee, Jungtaek Kim, Adam Kosiorek, Seungjin Choi, and Yee Whye Teh. 2019. Set Transformer: A Framework for Attention-based Permutation-Invariant Neural Networks. In *Proceedings of the 36th International Conference on Machine Learning*. PMLR, 3744–3753.

[12] Jin-Woong Lee, Woon Bae Park, Jin Hee Lee, Satendra Pal Singh, and Kee-Sun Sohn. 2020. A Deep-Learning Technique for Phase Identification in Multiphase Inorganic Compounds Using Synthetic XRD Powder Patterns. *Nature Communications* 11, 1 (2020), 86. doi:10.1038/s41467-019-13749-3

[13] Kangming Li, Kamal Choudhary, Brian DeCost, Michael Greenwood, and Jason Hattrick-Simpers. 2024. Efficient First Principles Based Modeling via Machine Learning: From Simple Representations to High Entropy Materials. *Journal of Materials Chemistry A* 12, 21 (2024), 12412–12422. doi:10.1039/D4TA00982G

[14] Kangming Li and Jason Hattrick-Simpers. 2024. *DFT Dataset for High Entropy Materials*. doi:10.5281/zenodo.10854500

[15] Freddie Markanday, Gareth Conduit, Bryce Conduit, Julia Purstl, Katerina Christofidou, Lova Chechik, Gavin Baxter, Christopher Heason, and Howard Stone. 2022. Design of a Ni-Based Superalloy for Laser Repair Applications Using Probabilistic Neural Network Identification. *Data-Centric Engineering* 3 (2022), e30. doi:10.1017/dce.2022.31

[16] Jonathon Markanday, Gareth Conduit, Bryce Conduit, Julia Purstl, Katerina Christofidou, Lova Chechik, Gavin Baxter, Christopher Heason, and Howard Stone. 2022. *Research Data Supporting Design of a Ni-Based Superalloy for Laser Repair Applications Using Probabilistic Neural Network Identification*. doi:10.17863/CAM.87406

[17] P. M. Mignanelli, N. G. Jones, E. J. Pickering, O. M. D. M. Messé, C. M. F. Rae, M. C. Hardy, and H. J. Stone. 2017. Gamma–Gamma Prime–Gamma Double Prime Dual-Superlattice Superalloys. *Scripta Materialia* 136 (2017), 136–140. doi:10.1016/j.scriptamat.2017.04.029

[18] D. B. Miracle and O. N. Senkov. 2017. A Critical Review of High Entropy Alloys and Related Concepts. *Acta Materialia* 122 (2017), 448–511. doi:10.1016/j.actamat.2016.08.081

[19] Felipe Oviedo, Zekun Ren, Shijing Sun, Charles Settens, Zhe Liu, Noor Titan Putri Hartono, Savitha Ramasamy, Brian L. DeCost, Siyu I. P. Tian, Giuseppe Romano, Aaron Gilad Kusne, and Tonio Buonassisi. 2019. Fast and Interpretable Classification of Small X-Ray Diffraction Datasets Using Data Augmentation and Deep Neural Networks. *npj Computational Materials* 5, 1 (2019), 60. doi:10.1038/s41524-019-0196-x

[20] Roger C. Reed. 2006. *The Superalloys: Fundamentals and Applications*. Cambridge University Press, Cambridge, UK. doi:10.1017/CBO9780511541285

[21] H. M. Rietveld. 1969. A Profile Refinement Method for Nuclear and Magnetic Structures. *Journal of Applied Crystallography* 2, 2 (1969), 65–71. doi:10.1107/S0021889869006558

[22] P. Thompson, D. E. Cox, and J. B. Hastings. 1987. Rietveld Refinement of Debye–Scherrer Synchrotron X-Ray Data from Al2O3. *Journal of Applied Crystallography* 20, 2 (1987), 79–83. doi:10.1107/S0021889887087090

[23] Pascal Marc Vecsei, Kenny Choo, Johan Chang, and Titus Neupert. 2019. Neural Network Based Classification of Crystal Symmetries from X-Ray Diffraction Patterns. *Physical Review B* 99, 24 (2019), 245120. doi:10.1103/PhysRevB.99.245120

[24] Anni Wang, Isabella Gallino, Sascha Sebastian Riegler, Yi-Ting Lin, Nishchay A. Isaac, Yesenia Haydee Sauni Camposano, Sebastian Matthes, Dominik Flock, Heiko O. Jacobs, Hung-Wei Yen, and Peter Schaaf. 2021. *DataSet / HEAF-NiAl: Ultrafast Formation of Single B2 Phase AlCoCrFeNi High Entropy Alloy Films by Combustion of a Reactive Ni/Al Multilayer as Heat Source*. doi:10.5281/zenodo.4623002

[25] Ruopeng Zhang, Shiteng Zhao, Jun Ding, Yan Chong, Tao Jia, Colin Ophus, Mark Asta, Robert O. Ritchie, and Andrew M. Minor. 2020. Short-Range Order and Its Impact on the CrCoNi Medium-Entropy Alloy. *Nature* 581, 7808 (2020), 283–287. doi:10.1038/s41586-020-2275-z

A Secondary Results and Diagnostics

The physics-constrained HEA experiment is exploratory and excluded from the formal matrix. Full per-seed results, parameter errors, instrument/overlap slices, and perturbation curves remain in machine-readable evidence. The PDF keeps the main four data figures in the body; secondary model-slice plots are released with the generated artifacts to preserve the page budget: `generated/hea_occupancy_signal_noise.png`, `generated/hea_perturbation_severity.png`, `generated/gamma_instrument_slices.png`, and `generated/gamma_overlap_difficulty.png`. Detailed generated tables remain in `generated/hea_sro_results.tex` and `generated/gamma_decomp_results.tex`.

A.1 Diagnostic Slices

The occupancy panel places every controlled pair below its parent-specific simulated-noise threshold, supporting abstention rather than forced ranking. The perturbation curve measures a different failure mode: SRO remains a defined structure-derived target even when observation quality degrades. For Gamma, instrument shift changes component F1 much less than severe overlap. Reporting these axes separately prevents a single average from hiding whether

failure is caused by observability, acquisition shift, or component coincidence.

A.2 Extended HEA-SRO Results

Across all four HEA splits, Spectrum Transformer is the strongest member of the formal model matrix. The exploratory physics-constrained variant changes MAE by only 0.0011–0.0024 relative to Spectrum Transformer, whereas ResNet1D can fall behind the composition-only control under composition and SRO shift. These small gains are why the physics-constrained variant is not promoted into the formal comparison.

A.3 Extended Gamma-Decomp Results

Peak Feature gives the strongest component F1 on both views and the strongest window count accuracy. The neural set predictors are less stable across seeds, and the physics-constrained model does not recover its full-pattern advantage on windows. Center, width, and area errors are therefore reported separately from component F1 rather than collapsed into a reconstruction-weighted score.